

# API vs MCP — Notes

Topic: Does MCP (Model Context Protocol) replace APIs? A full breakdown for developers, with side-by-side comparison + interview-ready points.

## Big takeaway: MCP does NOT replace APIs!

MCP servers still use APIs internally — MCP just adds an intelligent decision-making layer on top, built specifically for AI agents.

### 1. What is an API?

- Application Programming Interface — a bridge for communication between two software applications.
- Developer controls everything: which endpoint to call, what params to send, what response format to expect.
- Fast & predictable — but needs manual integration + docs for every service.

#### 5 HTTP methods:

- GET → fetch data
- POST → create new data
- PUT → insert / replace entire record
- PATCH → update a specific field
- DELETE → remove data

Why APIs succeeded: standardized protocols (HTTP/HTTPS, REST, GraphQL, gRPC), standard auth (OAuth, JWT, API keys), versioning, caching, error handling.

## 2. Where APIs fall short for AI

- Traditional APIs assume the DEVELOPER already knows which endpoint/params to use — everything is pre-defined.
- But AI doesn't follow fixed logic — it REASONS + decides dynamically.  
Example: "Check my calendar, find a free slot, schedule a meeting, then message me on Teams" → needs calendar API + Slack/Teams API + reasoning in between — one API call is not enough.
- So AI must discover tools, permissions, parameters & output formats at
- RUN TIME — plain APIs don't provide this automatically.
- Manually coding this for 100s of APIs → huge complexity + hard maintenance
- (different docs, auth, naming, error handling for every API).

## 3. Why MCP was created

- Anthropic introduced MCP to give AI a COMMON, STANDARDIZED way to discover and use tools/systems.
- Solves the inconsistency problem of every API being different.

## 4. Request Flow: API vs MCP

API → Developer writes code, sets endpoint/headers/auth, sends JSON, receives + parses response. Developer has FULL control.

MCP → User gives AI a task → AI THINKS what capability it needs → MCP server exposes available tools → AI picks the right tool → tool executes → result returns to AI → AI formats it for the user.

**API → Developer decides | MCP → AI decides**

## 5. Key Differences — Comparison Table

| Aspect                  | API                                      | MCP                                                                |
|-------------------------|------------------------------------------|--------------------------------------------------------------------|
| Decision-making         | Developer decides                        | AI decides                                                         |
| Tool/endpoint discovery | Manual                                   | Automatic                                                          |
| Speed                   | Faster for simple, direct tasks          | Slightly slower (extra reasoning) but smarter for multi-step tasks |
| Scalability             | Unwieldy at scale with many integrations | Scales better — new tools need no custom documentation             |
| Authentication          | API keys, OAuth, JWT, etc.               | Same auth methods — MCP shields the AI from raw credentials        |

## 6. Bottom-line Distinctions

- API = direct communication, developer-driven, manual integration  
→ best for traditional software.
- MCP = intelligent communication, AI-driven, automatic tool discovery  
→ best for AI-powered applications.